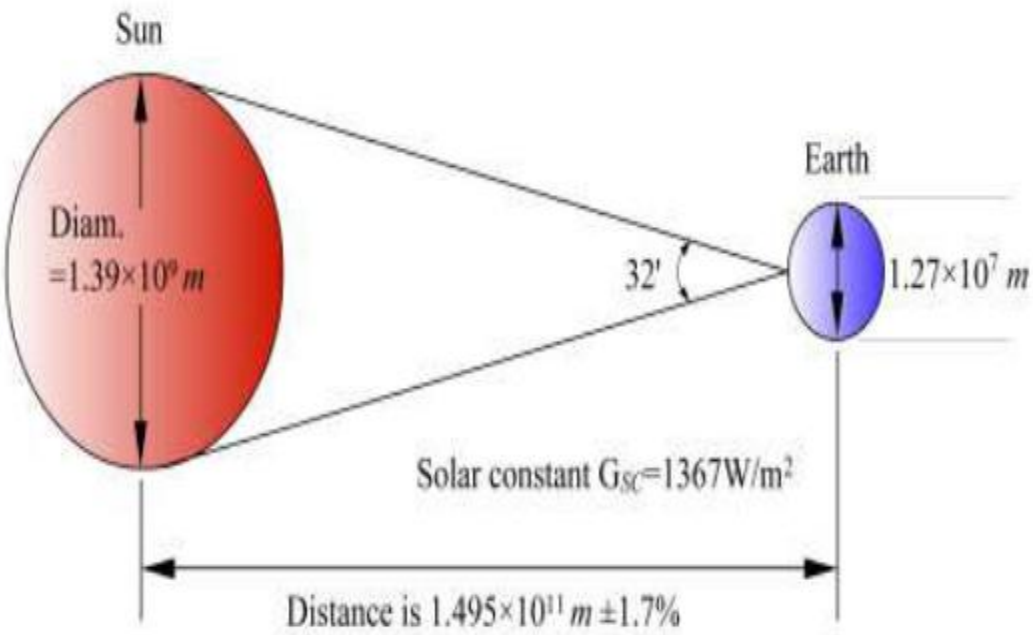
Solar Radiation Geometry

Solar Constant

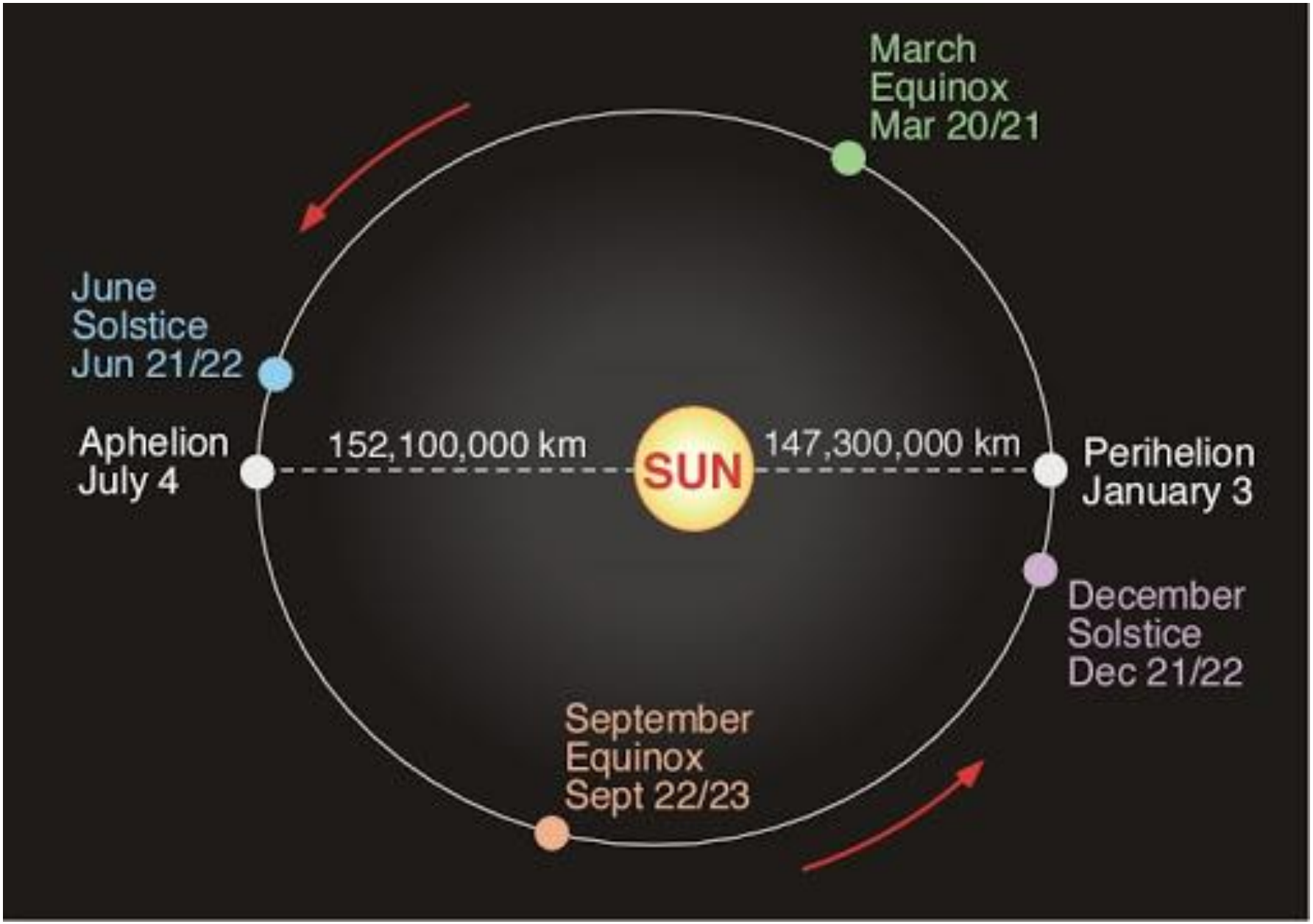
- Solar Constant is the rate at which energy is received from the sun on a unit area perpendicular to the rays of the sun at the mean distance of the earth from the sun.

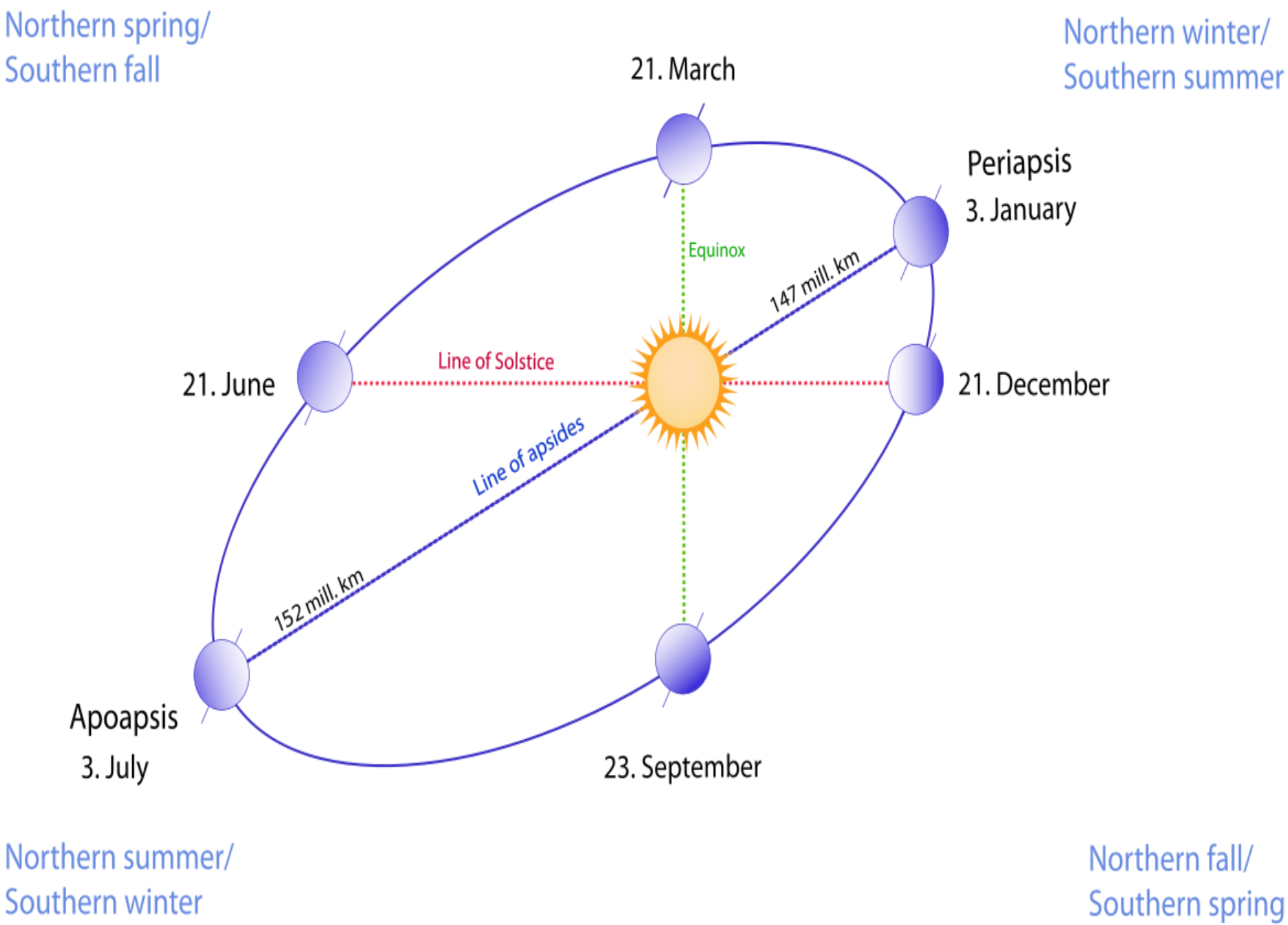
$$I_{sc} = 1367 \text{watt/m}^2$$

$$I'_{sc} = I_{sc}\left(1 + 0.033\cos\left[\frac{360}{365}n\right]\right.$$

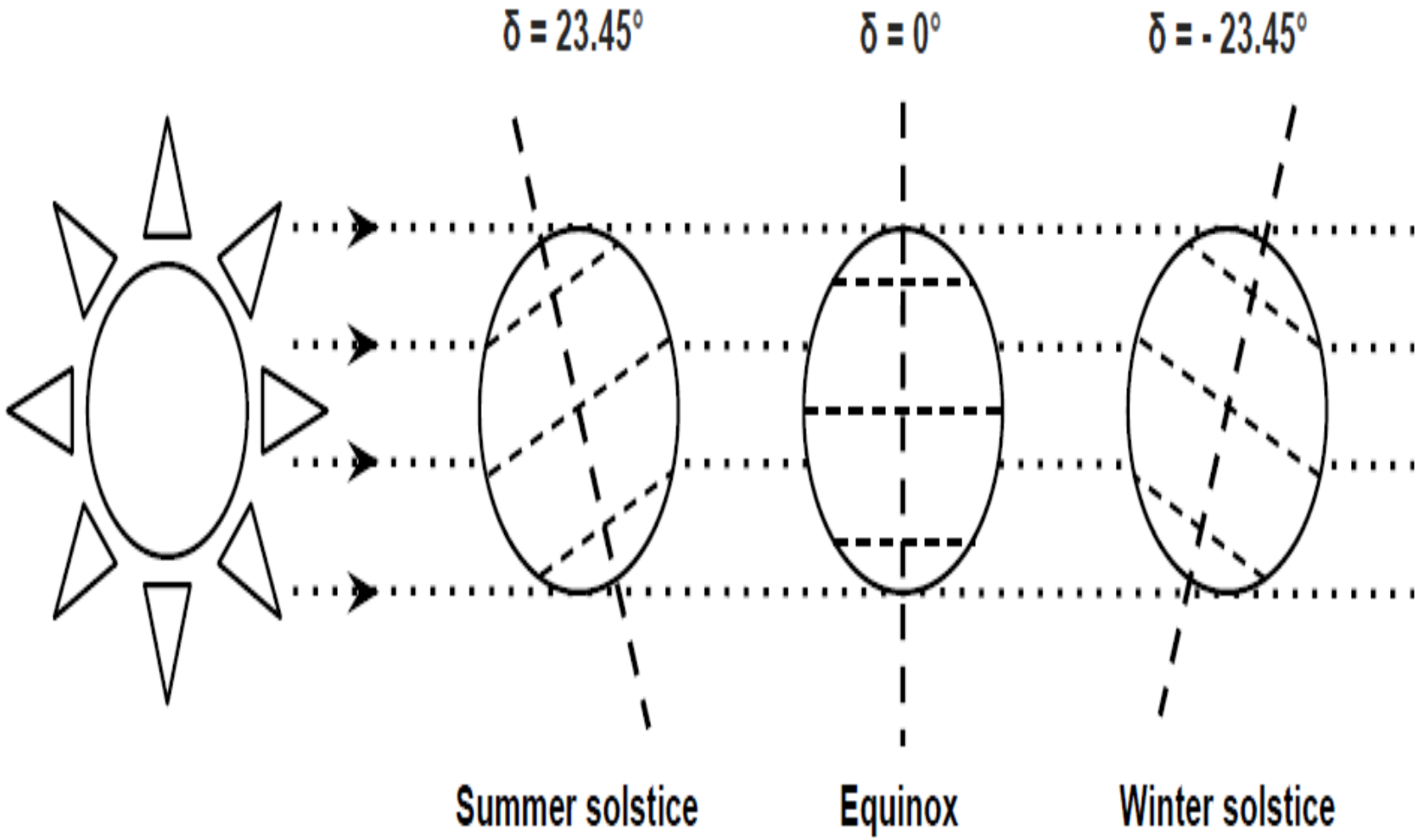


Solar Angle


March
Equinox
Mar 20/21
June
Solstice
Jun 21/22
Aphelion
July 4
152,100,000 km
SUN
147,300,000 km
Perihelion
January 3
December
Solstice
Dec 21/22
September
Equinox
Sept 22/23

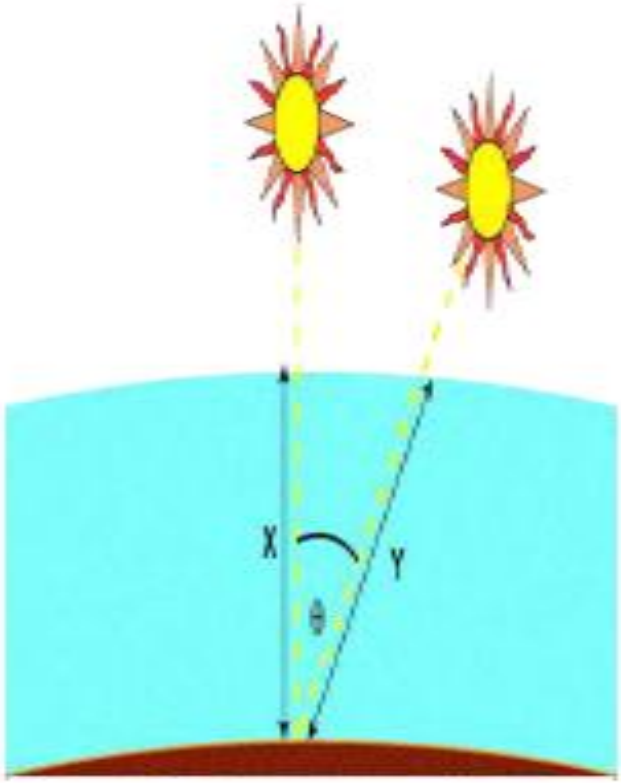

Northern spring/
Southern fall
21. March
Northern winter/
Southern summer
Periapsis
3. January
Equinox
147 mill. km
Line of Solstice
21. June
21. December
Line of apsides
152 mill. km
Apoapsis
3. July
23. September
Northern summer/
Southern winter
Northern fall/
Southern spring

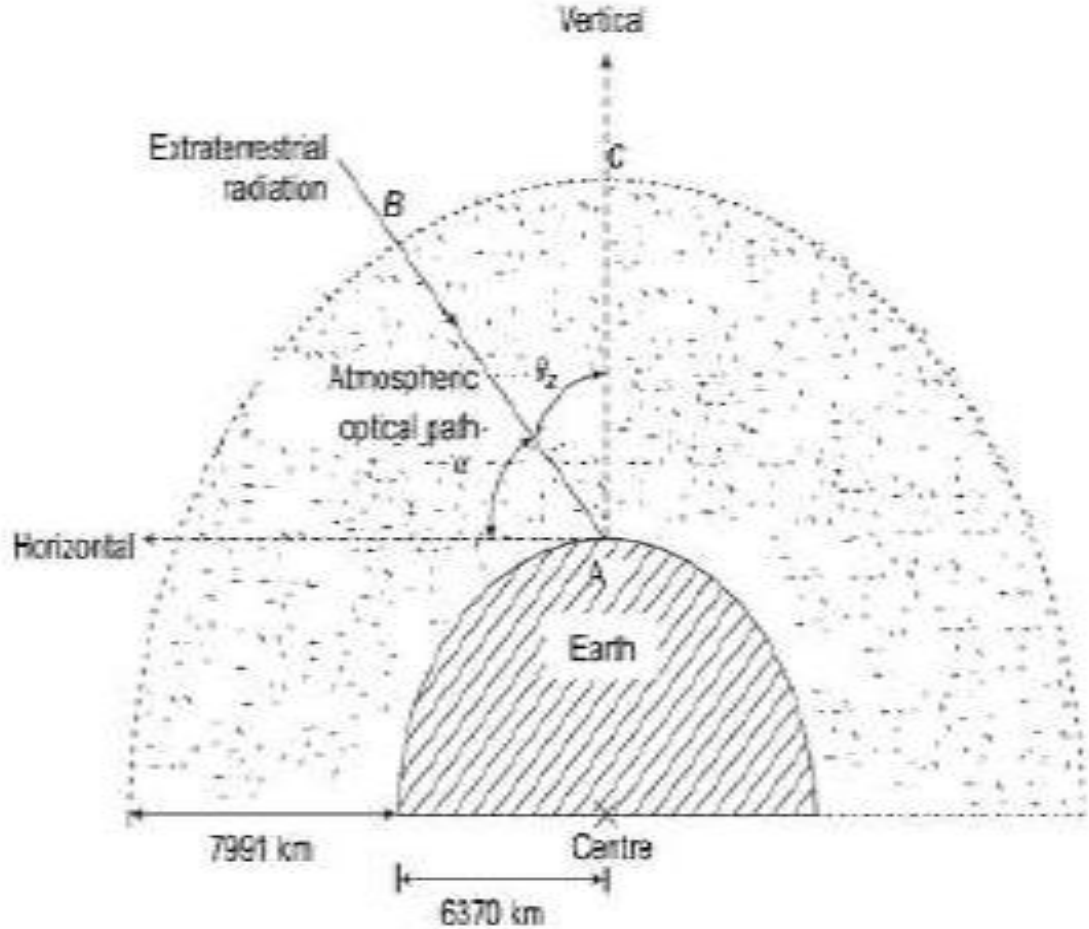
Ptolemaic view of sun motion



Air mass

- Air Mass', which is defined as the ratio of the path length through the atmosphere, which the solar beam actually traverses up to the ground to the vertical path length (which is minimum) through the atmosphere.
- Thus at sea level the air mass is unity when the sun is at the 'zenith' (highest position), i.e., when inclination angle is 90°.

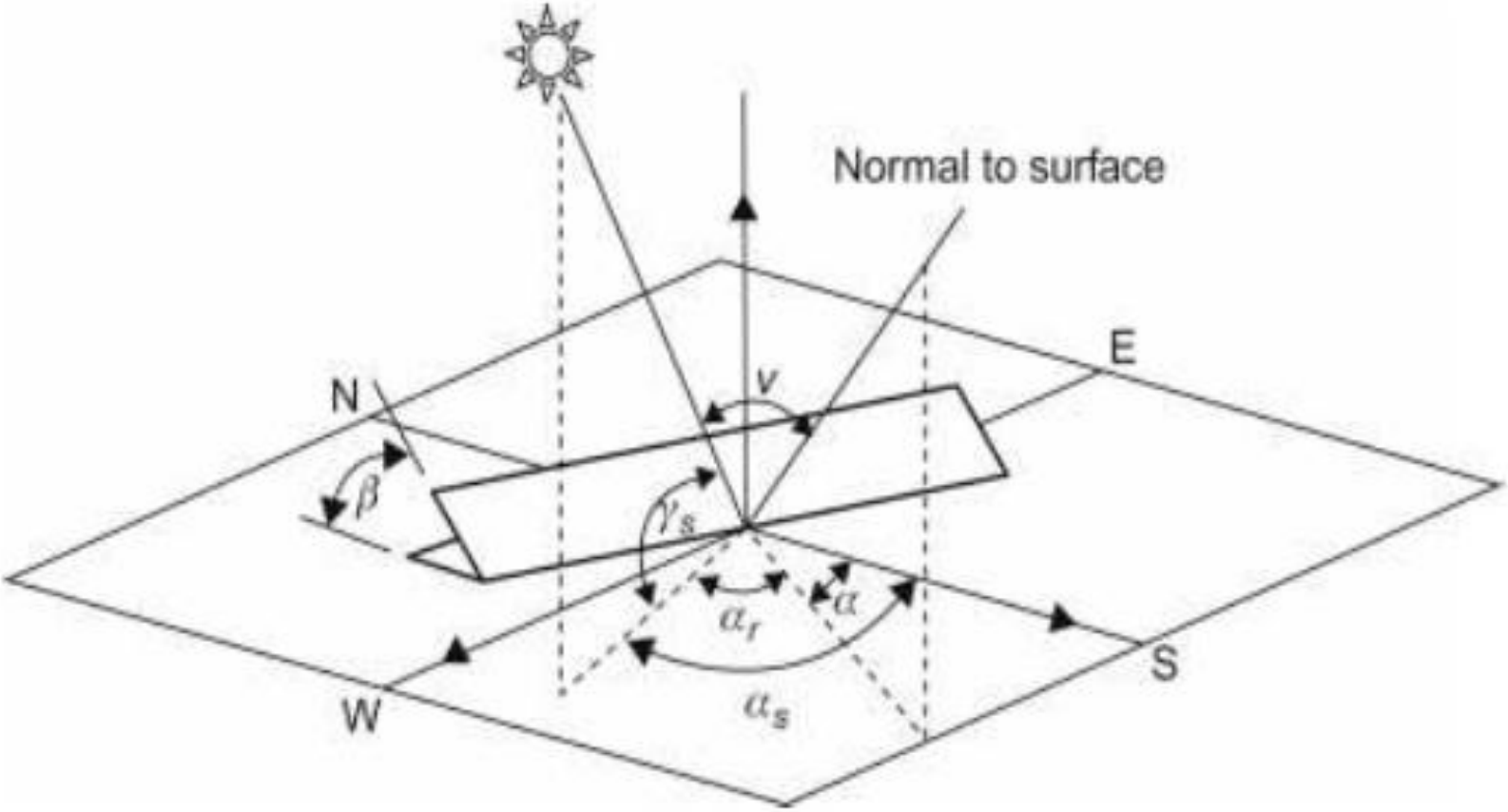




$$\text{airmass, } m = \frac{\text{path-length}_{\text{beamradiation}}}{\text{vertical-path-length}_{\text{atmosphere}}}$$

$$m = \frac{BA}{CA} = \frac{y}{x} = \sec\theta,\ m = [0, 2]$$

It is the angle between sun's ray incident on the plane surface (collector) and the normal to that surface.



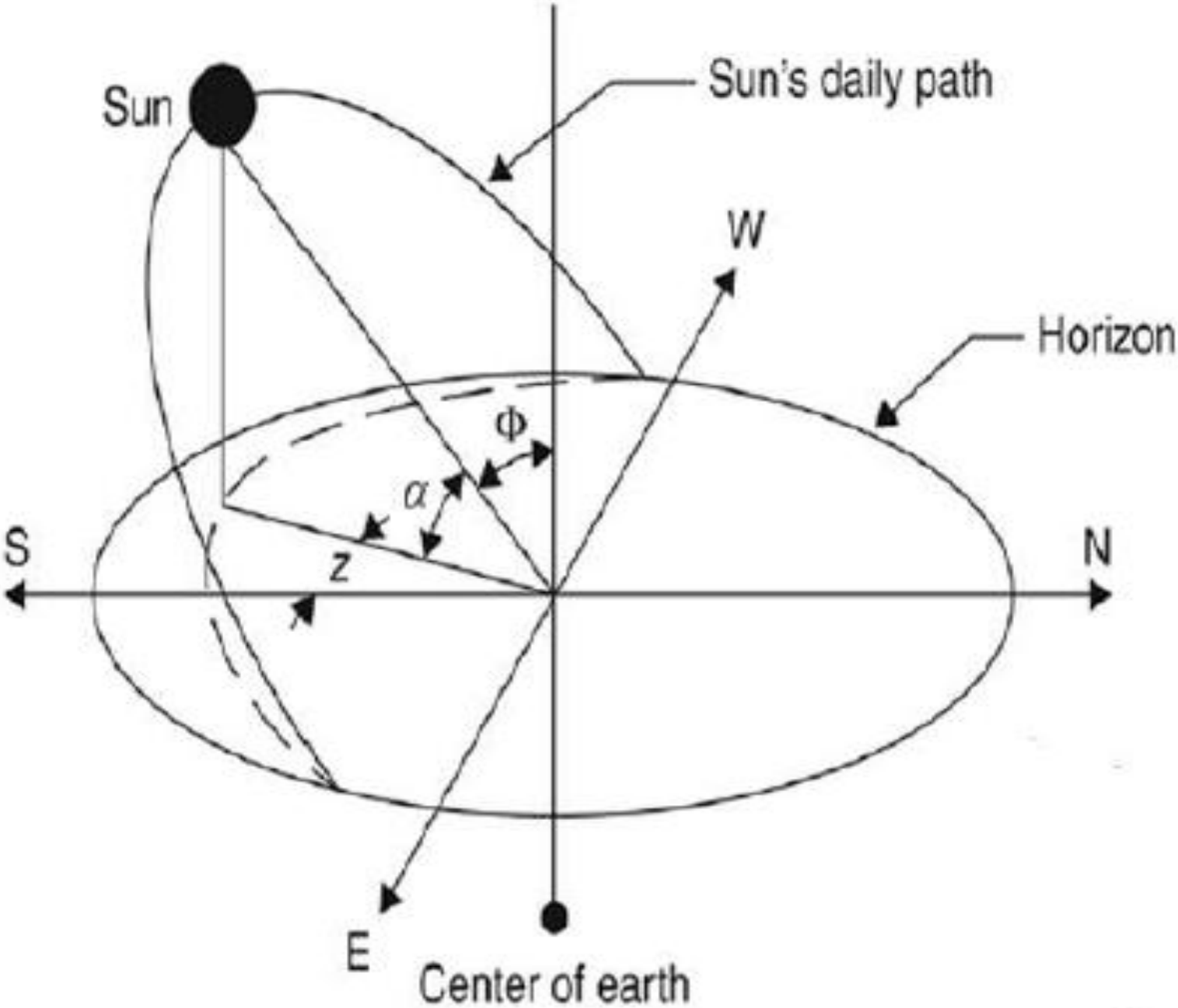
In general, the angle of incidence can be expressed as,

$$\cos\vartheta_1 = \begin{bmatrix} \cos\delta\cos\omega(\cos\phi\cos\beta + \sin\phi\sin\beta\cos\gamma) \\ +\cos\delta\sin\omega\sin\beta\sin\gamma \\ +\sin\delta(\sin\phi\cos\beta - \cos\phi\sin\beta\cos\gamma) \end{bmatrix}$$

1	$\beta = 0$	Horizontal surface
2	$\beta = 90$	Vertical surface
3	$\gamma = 0$	Inclined surface facing due south
4	$\gamma = 0, \beta = 90$	Vertical surface facing due south
5	$\gamma = 180$	Inclined surface facing due north
6	$\gamma = 180, \beta = 90$	Vertical surface facing due north

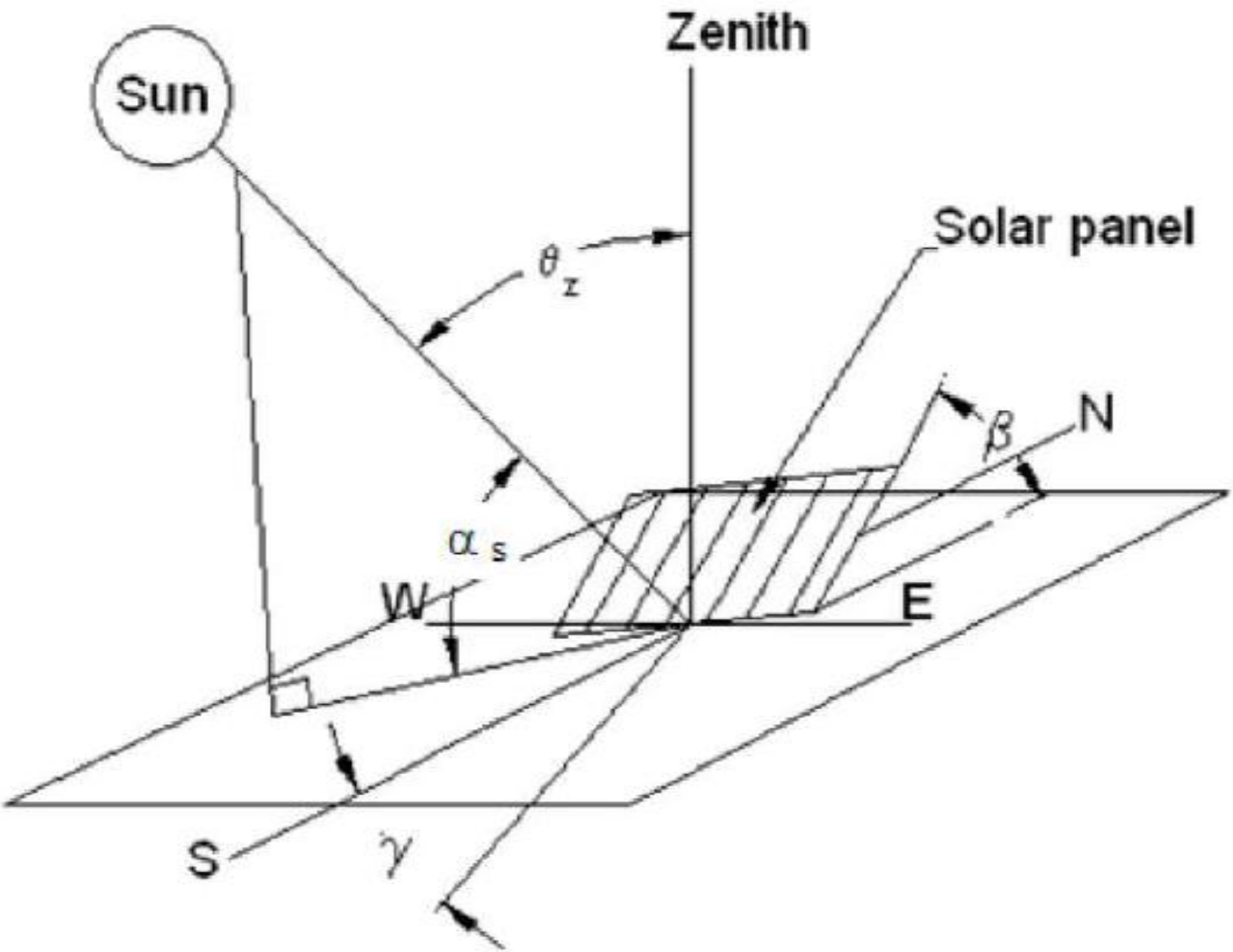
Solar inclination/altitude angle (α)

- The angle between sun's ray and its projection on horizontal surface is known as inclination angle.
- It is the angle of incidence between the beam radiation and normal to the plane surface/tilted surface/collector plate.


Sun
Sun's daily path
W
Horizon
Φ
α
S
z
N
E
Center of earth

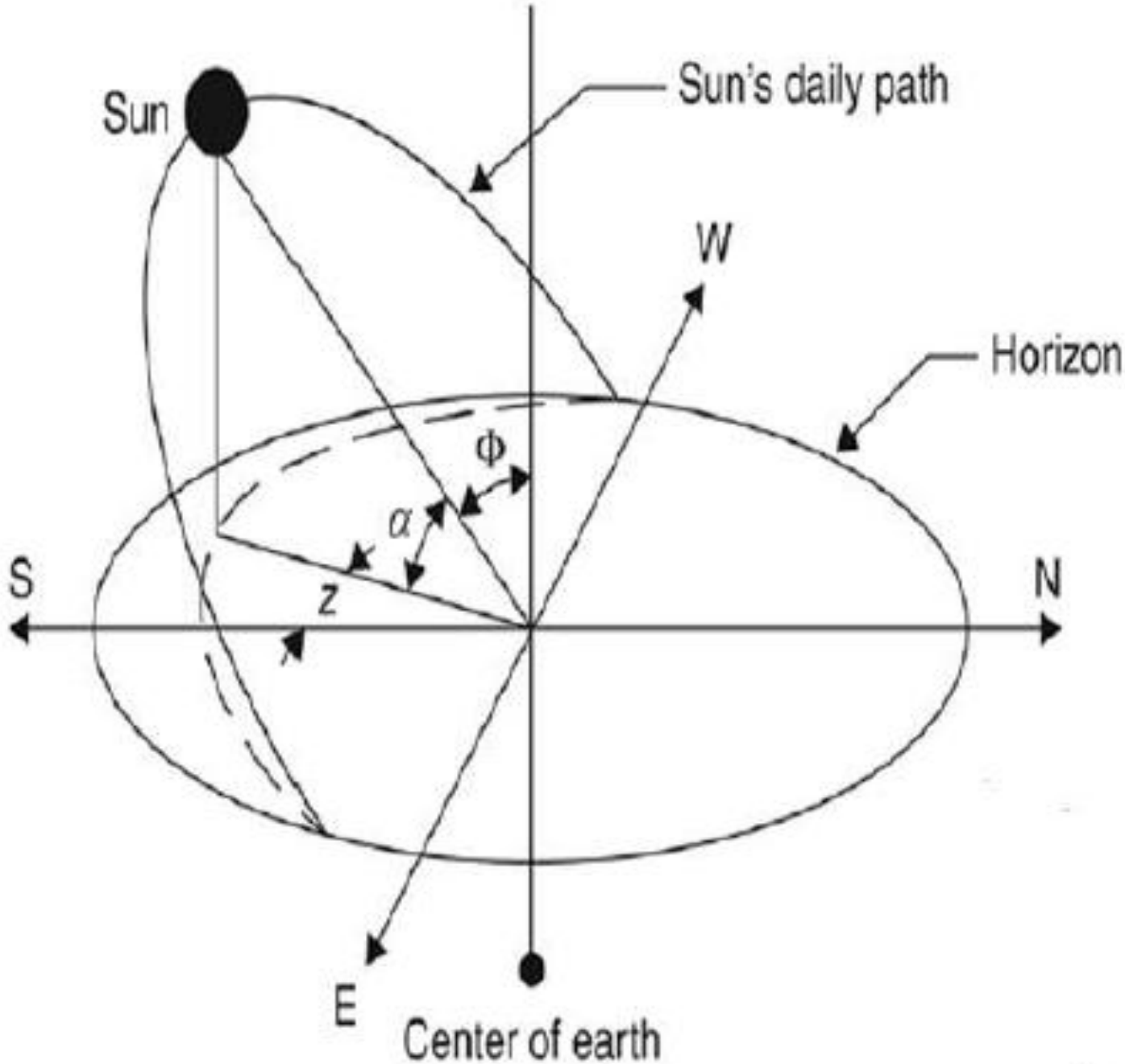
Slope or tilt angle (β)

- It is the angle made by the plane surface with the horizontal plane.
- It can vary from (0 to 180 degree)


Zenith
Sun
Solar panel
θ_z
β
N
α_s
W
E
S
γ

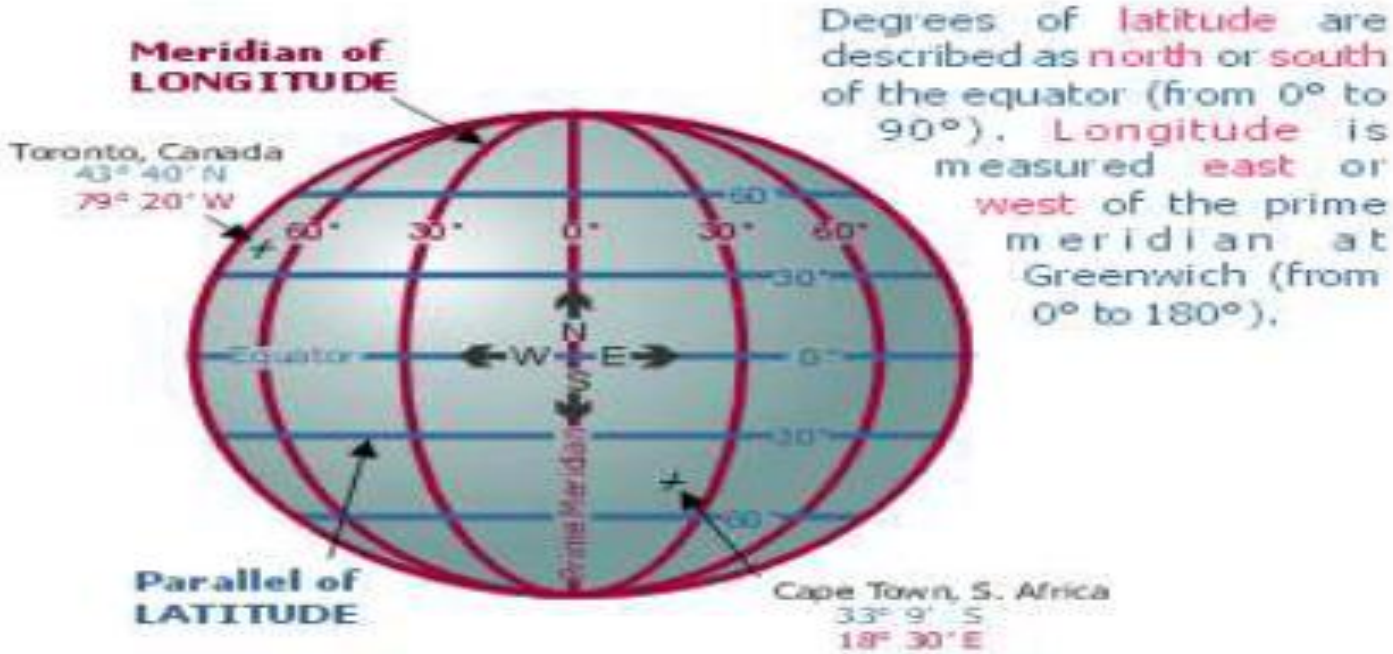
Zenith angle (ϑ_z)

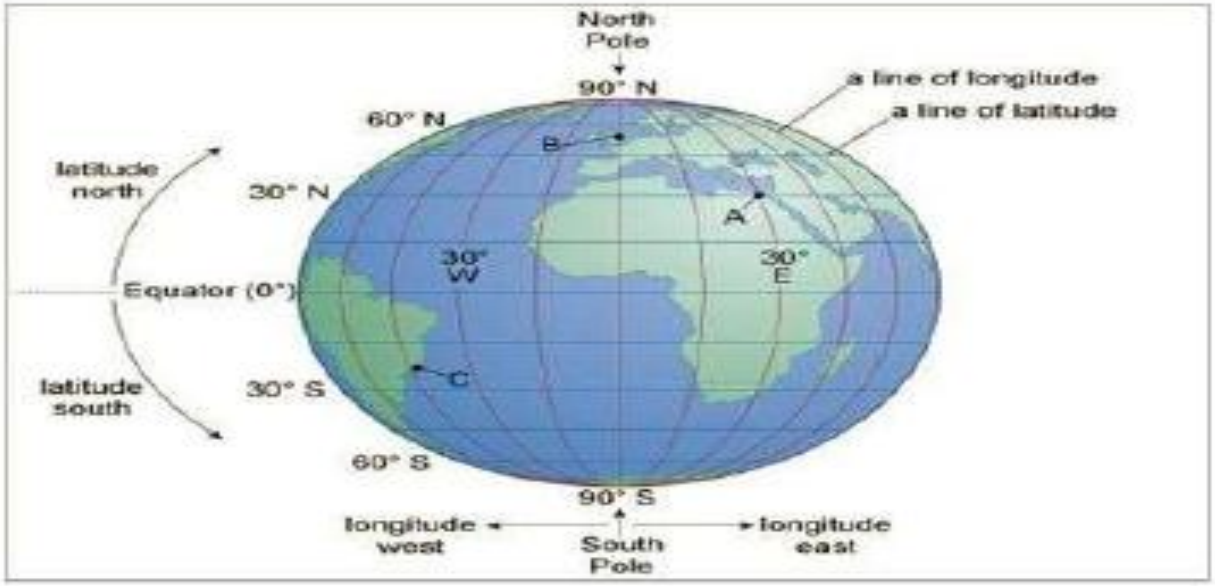
- The angle between sun's ray and its projection on horizontal surface is known as inclination angle


Sun
Sun's daily path
W
Horizon
Φ
α
S
z
N
E
Center of earth

Latitude angle (ϕ)

- The latitude of a location on earth's surface is the angle made by radial line, joining the given location to the center of the earth, with its projection on the equator plane as shown in Fig.
- The latitude is positive for northern hemisphere and negative for southern hemisphere.


Meridian of LONGITUDE
Toronto, Canada
43° 40' N
79° 20' W
60°
30°
0°
30°
60°
60°
30°
Equator
W
E
N
S
0°
30°
60°
Prime Meridian
Parallel of LATITUDE
Cape Town, S. Africa
33° 9' S
18° 30' E
Degrees of latitude are described as north or south of the equator (from 0° to 90°). Longitude is measured east or west of the prime meridian at Greenwich (from 0° to 180°).


North Pole
90° N
60° N
30° N
Equator (0°)
30° S
60° S
90° S
South Pole
latitude north
latitude south
longitude west
longitude east
30° W
30° E
A
B
C
a line of longitude
a line of latitude

Declination angle (δ)

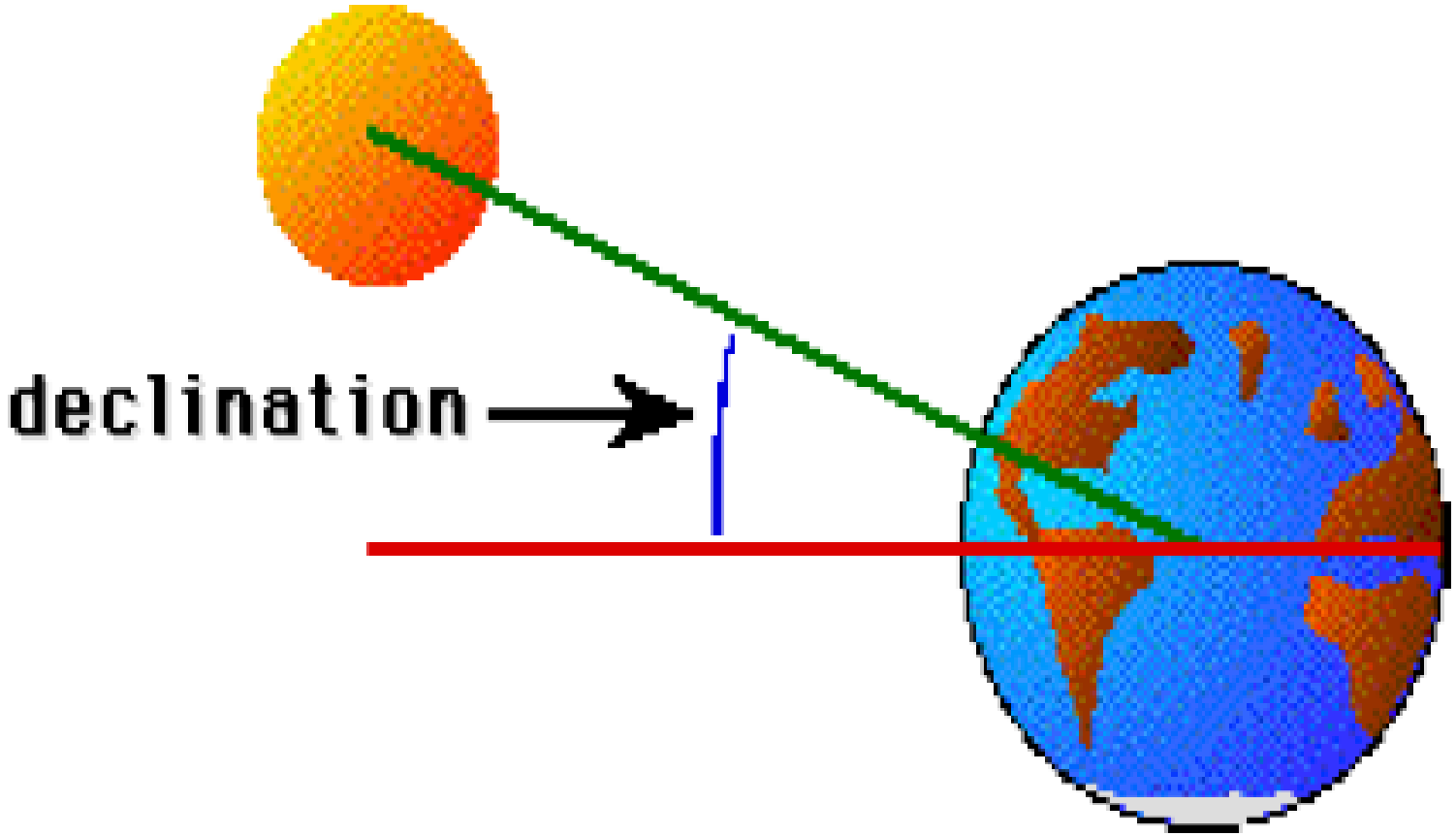
- It is defined as the angular displacement of the sun from the plane of earth's equator .
- It is positive when measured above equatorial plane in the northern hemisphere.

$$\delta = 23.45 \times \sin\left[\frac{360}{365}(284+n)\right] \text{degree}$$

$$\delta_{max} = 23.45(\text{June}-21),\ \delta_{min} = -23.45(\text{Dec}-21),$$

$$\delta = 0(\text{March}-21, \text{September}-22)$$

Where, n= day of the year counted from 1st January



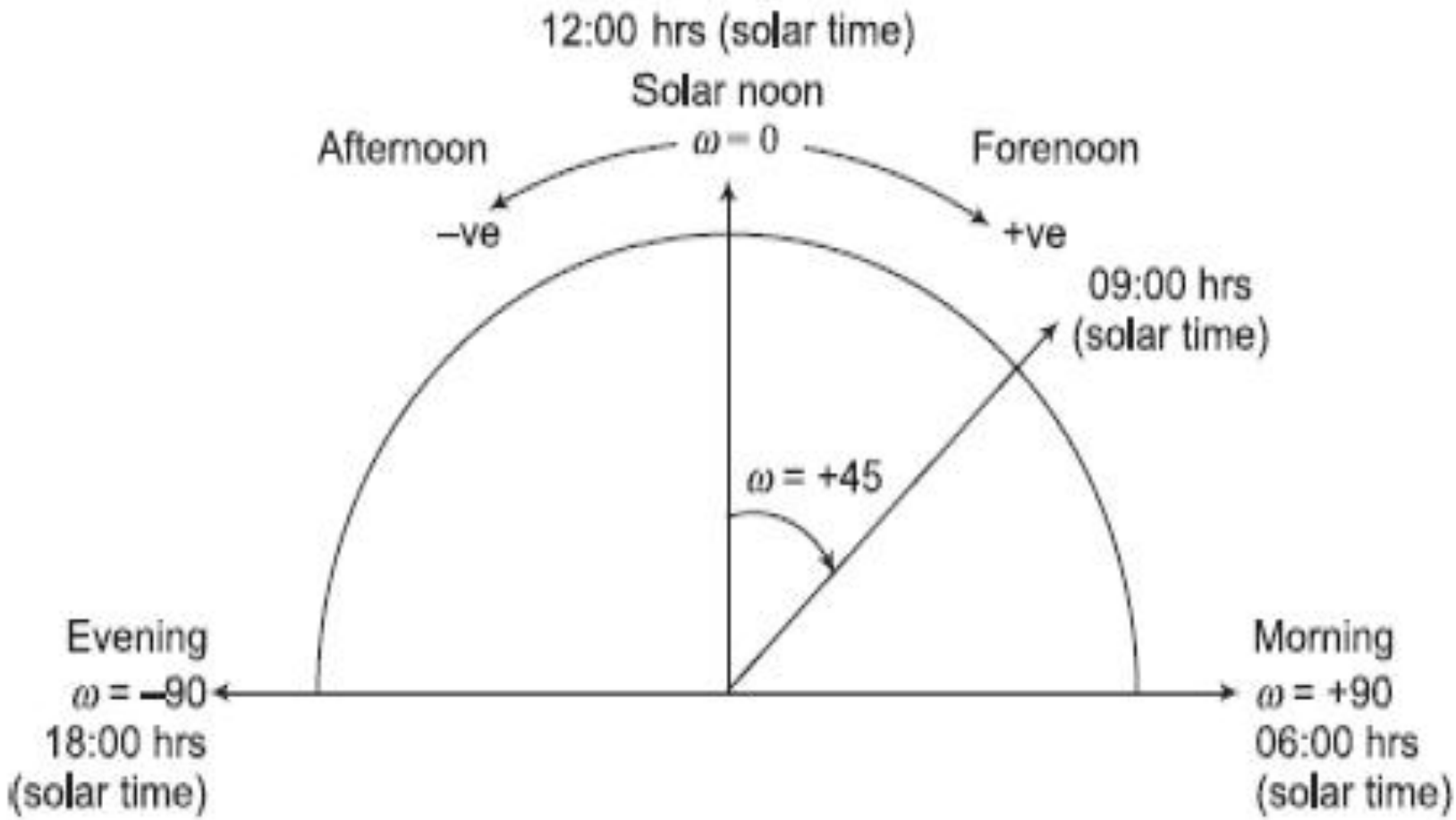
Hour angle (w)

- The hour angle at any moment is the angle through which the earth must turn to bring the meridian of the observer directly in line with sun's rays.

- It is an angular measurement of time from 180 degree and equivalent to 15 degree/hour. It varies from -180 degree to 180 degree.

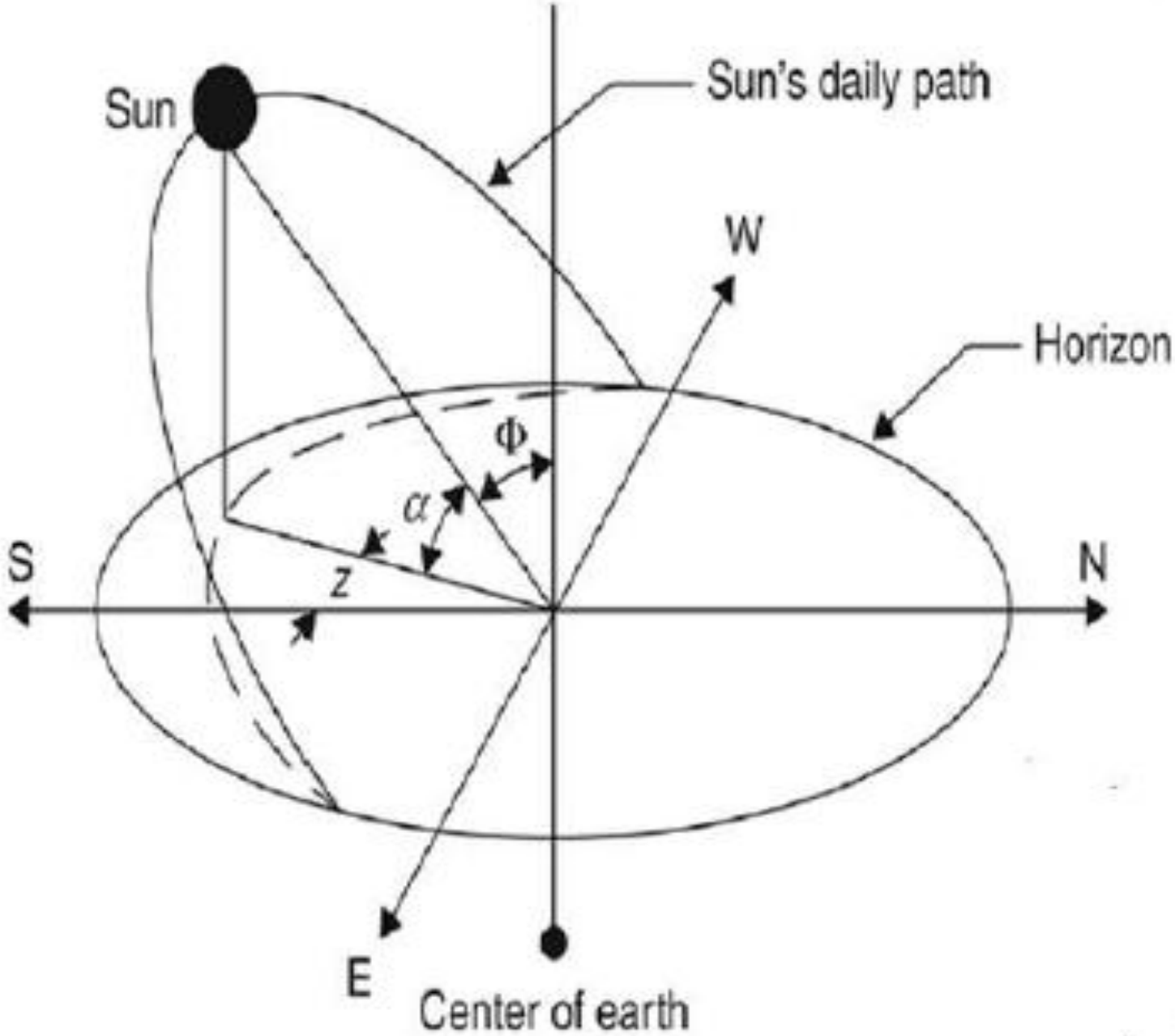
w=+ve, in morning
w=-ve, in afternoon

w=(180-(time x15))



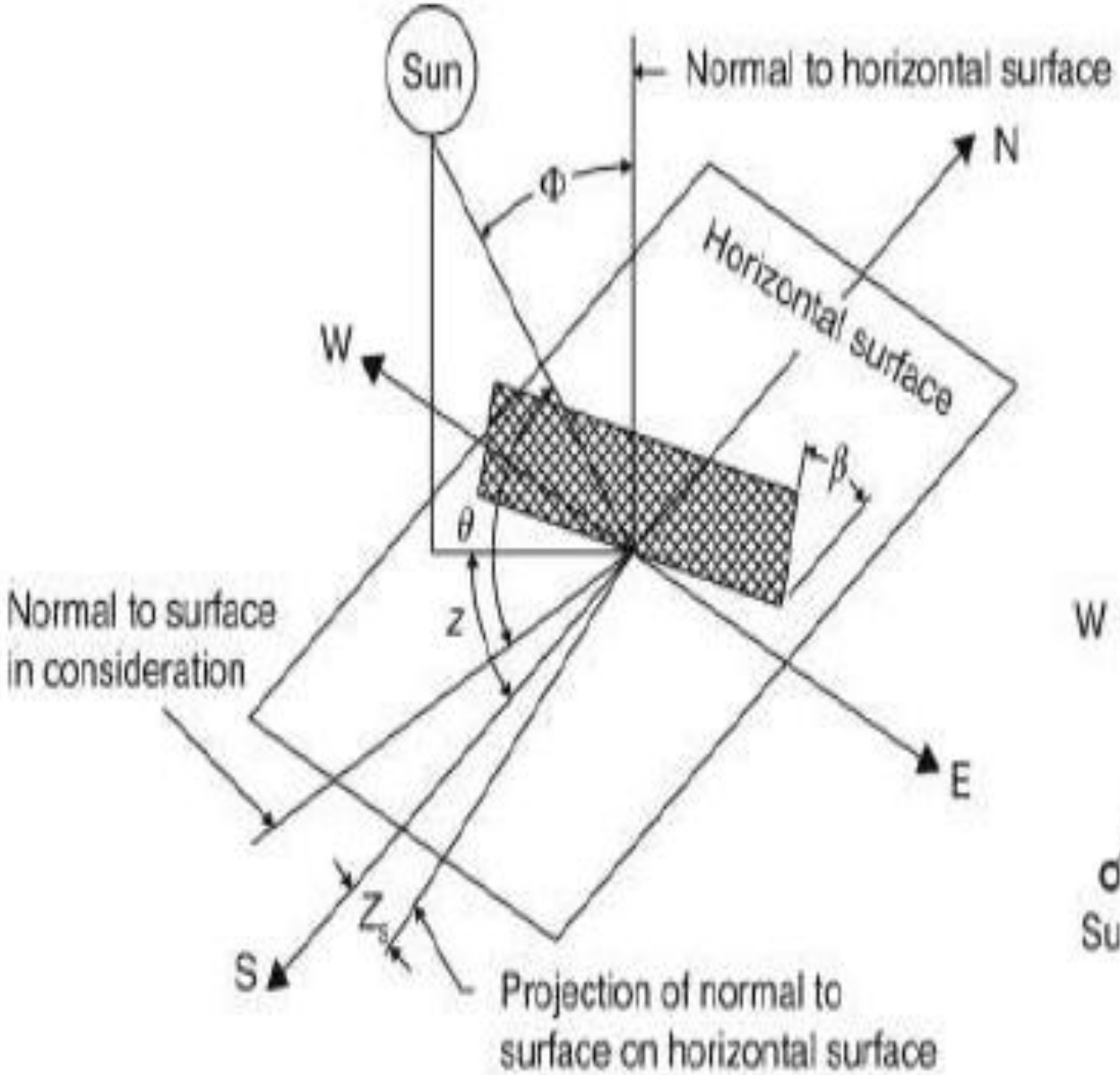
Solar Azimuthal angle

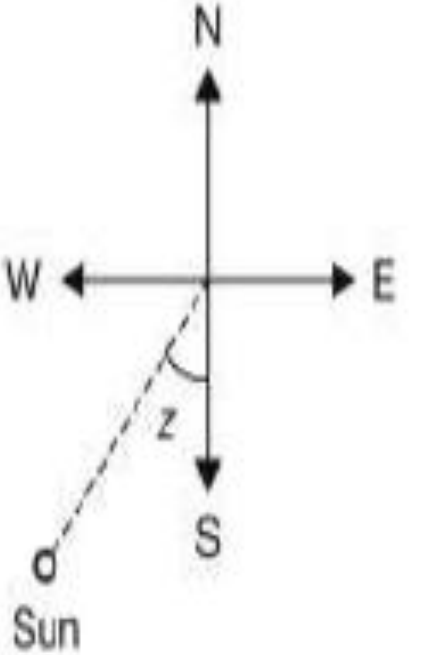
- It is the angle on a horizontal plane, between the line due south and the projection of sun's ray on the horizontal plane.
- It is taken as +ve when measured from south towards west.


Sun
Sun's daily path
W
Horizon
Φ
α
S
z
N
E
Center of earth

Surface Azimuthal Angle γ

- It is the angle in horizontal plane, between the line due south (OS) and the horizontal projection of normal to the inclined plane surface (OQ).
- It is taken as +ve when measured from south towards west.

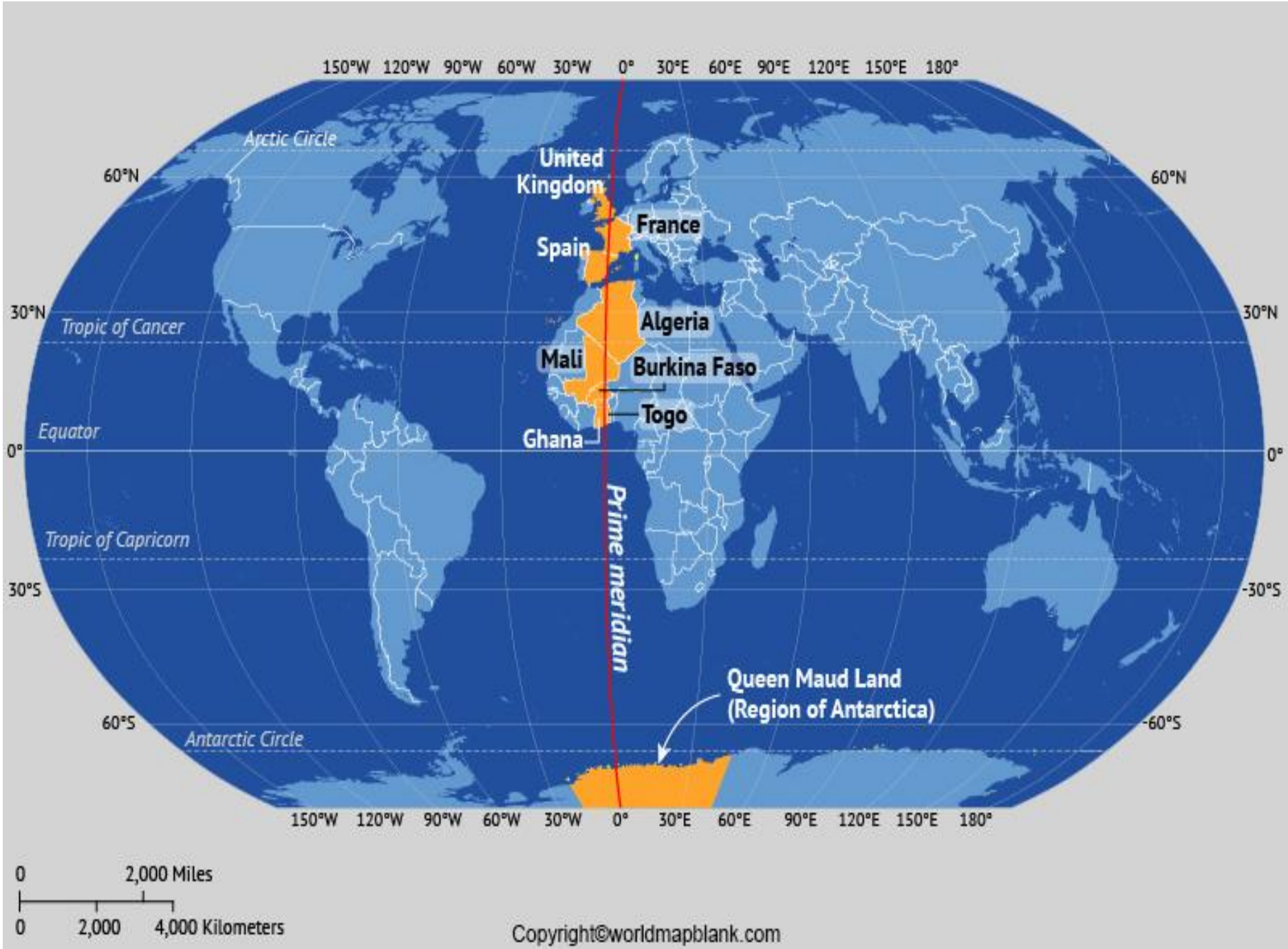




Plan view showing
solar azimuth angle

LAT

- The time used to calculate the hour angle is Local Apparent time (LAT).
- **Local apparent time** is that for the meridian of the observer.
- When the **apparent** (true) sun is (ahead of/behind) the mean sun, **apparent time** is (faster/slower) than mean civil **time**.
- The difference between **apparent** and civil **time** at any instant is called the equation of **time**.
- Solar time is measured with reference to solar noon, which is the time when the sun is crossing observer's meridian.
- At solar noon the sun is at the highest position in the sky. The sun traverses each degree of longitude in 4 minutes (as earth takes 24 hours to complete one revolution).


150°W 120°W 90°W 60°W 30°W 0° 30°E 60°E 90°E 120°E 150°E 180°
Arctic Circle
60°N
United Kingdom
France
Spain
30°N
Tropic of Cancer
Algeria
Mali
Burkina Faso
Togo
Equator
Ghana
0°
Prime meridian
Tropic of Capricorn
30°S
Queen Maud Land
(Region of Antarctica)
60°S
Antarctic Circle
0
2,000 Miles
0
2,000
4,000 Kilometers
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BASIS FOR COMPARISON	LOCAL TIME	STANDARD TIME
Meaning	Local time implies the time of a place determined on the basis of apparent movement of the sun.	Standard time refers to the fixed time for places falling in the same meridian, set in a country by law.
Variations	Changes continuously with the change in longitude.	Remains same for a particular country.
Longitude	Places on the same longitude have same local time.	Places on the same longitude have different standard time.
Reckoned by	Shadow cast by the sun.	Time zones

Local Apparent Time (LAT)

Local apparent time is that for the meridian of the observer.

When the **apparent** (true) sun is (ahead of/behind) the mean sun, **apparent time** is (faster/slower) than mean civil **time**.

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LAT

LAT=

$$\text{Standard}_{\text{Time}} +/- 4\text{min } (\text{Standard}_{\text{Timelongitude}} - \text{Longitude of location}) + E$$

E=Time Correction Factor
+ve=western hemisphere
−ve=eastern hemisphere

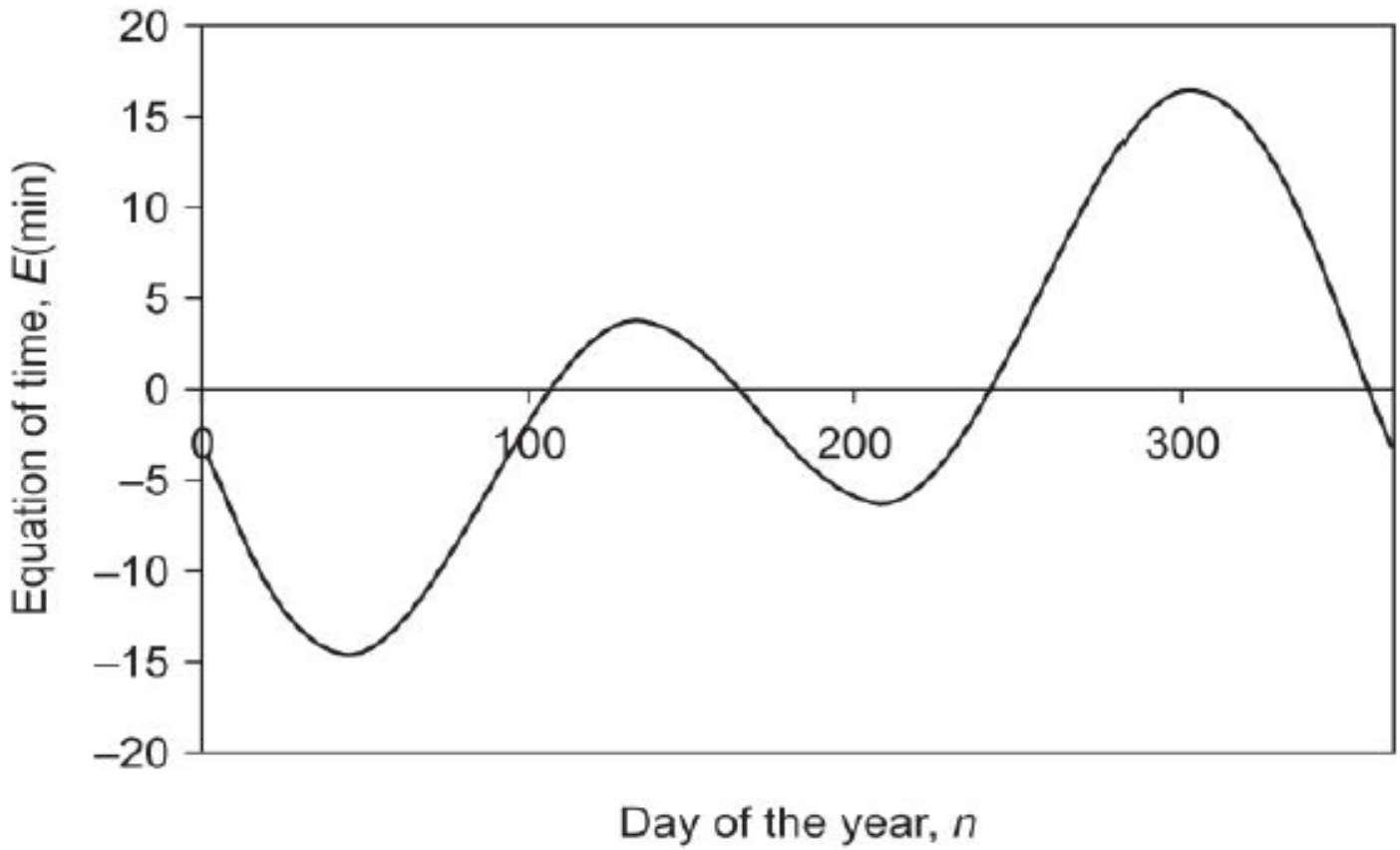
- E is the correction arising out of the variation in the length of the solar day due to variations in earth's rotation and orbital revolution, and is called equation of time.
- E can be determined either by calculation using following equation (within an accuracy of half a minute) or from the chart given

Time correction Factor

$$E = 9.87 \sin 2B - 7.53 \cos B - 1.5 \sin B \text{ min.}$$

where $B = (360/364)(n - 81)$

and n = day of the year, starting from 1st January



LAT=Standard Time+/-4^{min} (Standard $Time_{longitude}$-Longitude of location) +Time correction

+ve=western hemisphere

-ve=eastern hemisphere